

ON THE FAILURE OF BOUNDARY-SEEKING DISTILLATION IN BOTTLENECKED GENERATIVE ARCHITECTURES

Mohamed Amine Kina

Universität Bremen
mkina@uni-bremen.de

ABSTRACT

Data-free knowledge distillation transfers a teacher model to a student without access to the original training data. Contrastive Abductive Knowledge Extraction (CAKE) achieves this for classifiers by synthesizing samples near the teacher’s decision boundary. We investigate whether this boundary-seeking principle extends to autoencoder distillation. To enable a direct comparison, we reformulate continuous reconstruction as a dense, per-feature classification task, allowing the decoder to output categorical logits. We show that boundary-seeking objectives are fundamentally ill-posed in bottlenecked generative architectures. CAKE operates on a single, instance-level objective, but a decoder acts as an array of tightly coupled, feature-level classifiers constrained by a shared low-dimensional bottleneck. Independently sampling contrastive targets for these coupled outputs violates the geometry of the learned latent manifold and produces severe gradient conflicts instead of informative boundary samples. Manifold-aware synthesis — specifically, noise propagation through the teacher — bypasses these conflicts entirely and establishes an effective baseline for data-free generative distillation.

1. INTRODUCTION

The increasing availability of pre-trained models across machine learning has made knowledge distillation an attractive tool for model compression, continual learning, and adaptation. Traditionally, however, distillation assumes access to the original training data — an assumption that is frequently violated in practice due to privacy constraints, proprietary data, or simple unavailability. Contrastive Abductive Knowledge Extraction (CAKE) [1] was proposed precisely to address this gap, enabling data-free knowledge distillation in a model-agnostic manner by generating synthetic samples that probe a teacher model’s decision boundary through a contrastive diffusion process.

Despite its efficacy in discriminative tasks, the mechanism remains fundamentally dependent on the presence of a classification-based decision surface to guide the diffusion process. This raises a natural question: can CAKE be ex-

tended beyond classification to unsupervised generative models such as autoencoders? In this work, we investigate this question by reformulating standard continuous reconstruction into a dense, feature-level classification task, allowing us to apply CAKE’s contrastive framework directly to a bottleneck-constrained architecture.

Through this setup, we make two primary contributions. First, we provide a theoretical and empirical argument for why CAKE fundamentally fails in the autoencoder setting, tracing the optimization collapse to the strict constraints imposed by the shared learned latent manifold. Second, we propose and analyze iterative noise propagation as a simpler, intuitive, and highly effective alternative mechanism for data-free knowledge distillation from autoencoders.

2. BACKGROUND

2.1. Knowledge Distillation and CAKE

Knowledge distillation [2] aims to transfer the knowledge encoded in a teacher model f^T into a typically smaller student model f^S , by training the student to mimic the teacher’s outputs rather than learning from scratch. In the supervised classification setting, this is achieved by minimising a combination of the cross-entropy loss on ground-truth labels and a softer matching loss on the teacher’s output distribution, controlled by a temperature parameter τ .

CAKE [1] extends this to the data-free setting by removing the requirement for original training data entirely. Rather than approximating the data distribution, CAKE targets the teacher’s decision boundary directly — what the authors term *abductive knowledge*. It generates pairs of synthetic samples assigned to different classes and applies a contrastive loss to pull them toward opposing sides of the decision boundary. Stochastic noise induced by SGD’s decaying step-size schedule then sweeps the samples along the boundary to ensure broad coverage. The result is a synthetic dataset that is sufficient to train a capable student classifier.

Decision Boundaries Stack

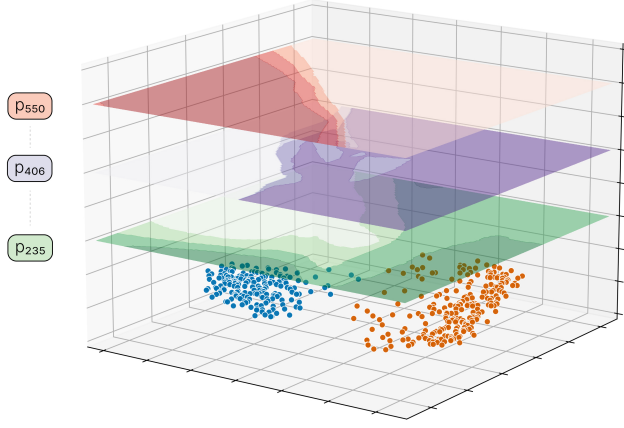


Fig. 1: Visualization of pixel-level decision boundaries stacked over the shared latent space. The clusters represent MNIST digits 1 (blue) and 0 (orange). The left panel displays the distinct decision surfaces for three selected pixels (p_{235} , p_{406} , and p_{550})

2.2. Autoencoders

An autoencoder is an unsupervised neural network composed of an encoder q_ϕ and a decoder p_θ . Given an input $\mathbf{x}$, the encoder maps it to a lower-dimensional latent representation $\mathbf{z} = q_\phi(\mathbf{x})$, and the decoder reconstructs the input as $\hat{\mathbf{x}} = p_\theta(\mathbf{z})$. Training minimises a reconstruction loss, typically the mean squared error:

$$\mathcal{L}_{AE}(\mathbf{x}) = \|\mathbf{x} - p_\theta(q_\phi(\mathbf{x}))\|^2$$

The bottleneck — the low-dimensional latent space $\mathbf{z}$ — is the defining feature of this architecture. By forcing all information through a compressed representation, the encoder learns to retain only the structure most useful for reconstruction, effectively learning a manifold that approximates the support of the training data distribution.

3. CAKE’S INAPPLICABILITY TO AUTOENCODERS

3.1. Adapting CAKE to the Autoencoder Setting

CAKE is naturally defined for classifiers, where the teacher maps an input to a single global class distribution. To test whether the same mechanism can be transferred to autoencoders, we reformulate the MNIST autoencoder as a pixel-wise classification model. Instead of reconstructing continuous grayscale values, each pixel is assigned to one of four intensity bins, corresponding to the background and three foreground levels. The decoder therefore outputs:

$$f_\theta(\mathbf{z}) \in R^{784 \times 4}$$

where each of the 784 pixels has an independent four-class logit vector. Given an input image $\mathbf{x}$, the encoder produces a 12-dimensional latent code:

$$\mathbf{z} = g_\phi(\mathbf{x})$$

This construction makes the autoencoder superficially compatible with CAKE: one can sample a target class map $\mathbf{y}$ and optimize a synthetic input $\mathbf{x}$ to probe the teacher’s decision boundaries. Following the standard CAKE formulation, the full extraction objective combines three components: a target classification loss, a contrastive loss to diffuse sample pairs toward the boundary, and a Total Variation (TV) prior:

$$\mathcal{L}(\mathbf{x}, \mathbf{y}) = \lambda_{cls} \mathcal{L}_{cls}(\mathbf{x}, \mathbf{y}) + \lambda_{contr} \mathcal{L}_{contr} + \lambda_{TV} \mathcal{L}_{TV}(\mathbf{x})$$

To adapt this for the autoencoder, the classification component $\mathcal{L}_{cls}$ must be evaluated densely across the output space:

$$\mathcal{L}_{cls}(\mathbf{x}, \mathbf{y}) = \sum_{j=1}^{784} \text{CE}(f_{\theta,j}(g_\phi(\mathbf{x})), y_j)$$

where $y_j \in \{1, \dots, 4\}$ is the independently sampled target class for pixel j .

However, this objective reveals the central mismatch between CAKE and autoencoder distillation. In CAKE for standard classifiers, a synthetic sample is optimized toward one global class label. In the autoencoder case, a single 12-dimensional bottleneck must satisfy 784 pixel-level labels simultaneously. This exposes the fundamental architectural flaw of applying boundary-seeking diffusion to dense generative models: independently sampling targets for highly coupled output features guarantees severe gradient conflicts across the shared latent manifold.

3.2. Latent Manifold Constraint

The decoder does not represent all possible 28×28 four-class images. Its outputs are constrained by the 12-dimensional latent bottleneck learned from MNIST. Thus, the set of possible reconstructions is a small structured subset of the full combinatorial output space:

$$\mathcal{M}_\theta = \{\arg \max f_\theta(\mathbf{z}) : \mathbf{z} \in R^{12}\} \subset \{1, \dots, 4\}^{784}.$$

When a contrastive distillation method like CAKE generates a target class map $\mathbf{y}$ by independently sampling or perturbing pixel-wise classes, the resulting target is drawn from 4^{784} possible combinations. Because these independent pixel targets are unconstrained by the underlying data distribution, they almost never correspond to a valid MNIST-like geometry. Consequently, they are overwhelmingly likely to fall outside the decoder’s learned manifold $\mathcal{M}_\theta$.

Figure 1 illustrates this constraint geometrically. The bottom layer shows a 2D PCA projection of encoded MNIST

digits. The stacked contour planes show the decoder’s decision regions for selected individual pixels. A single point in latent space determines the output class for all pixels simultaneously.

3.3. Gradient Conflict Under Random Pixel Targets

The failure of CAKE in this setting follows directly from this coupling. When targets are sampled independently for all pixels, different pixels typically demand incompatible movements in the latent space. Applying the chain rule, the gradient of the dense classification objective with respect to the shared bottleneck is a sum of 784 terms:

$$\nabla_{\mathbf{z}} \mathcal{L}_{cls} = \sum_{j=1}^{784} \nabla_{\mathbf{z}} \text{CE}(f_{\theta,j}(\mathbf{z}), y_j)$$

Each term attempts to move the same latent code toward a region where one specific pixel achieves its assigned target class. Since the targets are sampled independently, these directional vectors rarely agree. Satisfying the target for one pixel routinely moves the latent code away from the target region required by another pixel.

Figure 2 visualizes this gradient conflict for two independent pixels (pixel₅₅₀ and pixel₄₀₆) using mathematically derived gradient vector fields. Because both pixels are conditioned on the same input, their shared latent embedding z (blue dot) occupies the exact same coordinate space. To satisfy the independently sampled target for pixel₅₅₀ (class 3), the gradient field demands a shift upward and to the left. Simultaneously, satisfying the target for pixel₄₀₆ (class 0) demands a contradictory shift to the right. Rather than discovering informative teacher boundaries, this severe gradient conflict tears the optimization in opposing directions, pushing the latent code toward unstable or off-manifold regions.

This is not merely an artifact of the 2D PCA visualization. In the full 12-dimensional latent space, each pixel decision boundary is an 11-dimensional hypersurface. The same conflict remains: a single low-dimensional latent code is being asked to satisfy hundreds of independently sampled categorical constraints. The problem is therefore massively over-constrained. There are only 12 latent degrees of freedom, but 784 strictly enforced pixel-wise targets.

3.4. Why This Differs from Standard CAKE

This structural mismatch explains why CAKE succeeds in settings such as the Two-Moons dataset or standard image classification, but does not transfer naturally to autoencoders. In a standard classifier, the model maps an input to one unified, global label. CAKE only needs to move a synthetic sample toward a single decision boundary separating discrete global classes. Because the optimization is guided by a singular target, the high-dimensional input space provides more than enough degrees of freedom to satisfy it.

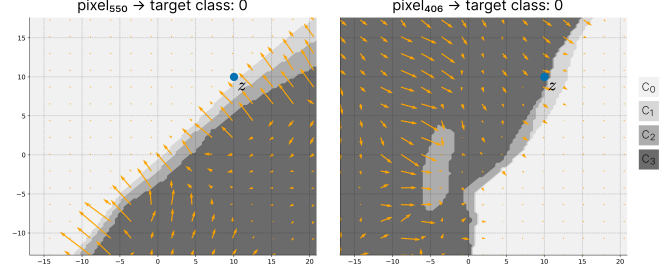


Fig. 2: Gradient vector fields demonstrating latent coupling. The shared latent embedding z is pulled upward and to the left to satisfy the target for pixel₅₅₀ (left), but pulled to the right to satisfy pixel₄₀₆ (right). This opposing directional constraint illustrates why independently sampled targets fail in bottlenecked architectures.

The autoencoder’s decoder has the exact opposite structure. It maps a highly constrained, low-dimensional latent code to a high-dimensional, structured output. The target is not one class label, but an entire image-sized grid of independent pixel classes. Randomly assigning these pixel classes ignores the geometric correlations learned by the decoder: strokes are spatially coherent, neighboring pixels are highly dependent, and digit identity is represented globally. Consequently, almost all randomly generated target configurations represent off-manifold noise that cannot be produced by any latent code.

4. EXPERIMENTAL SETUP AND RESULTS

To empirically validate the structural limitations of boundary-seeking optimization and evaluate alternative data-free extraction methods, we construct a controlled distillation pipeline. We first train a teacher autoencoder on the four-class pixel-wise MNIST task to full convergence. We then synthesize a transfer batch using various extraction strategies without accessing the original training data. Finally, a student autoencoder—architecturally identical to the teacher but with half the parameter count—is trained from scratch using the synthesized batch.

4.1. Evaluation Metrics

Because the autoencoder reconstruction is formulated as a dense classification task, standard image quality metrics (such as MSE or PSNR) are inapplicable. Instead, we evaluate the student’s fidelity to the teacher using strict categorical and spatial overlap metrics:

- **Overall Accuracy:** The global proportion of pixels across the entire test set where the student’s predicted class perfectly matches the teacher’s prediction.

- **Foreground mIoU (F-mIoU):** Our primary benchmark metric. It averages the Jaccard Index of the three foreground classes ($c \in \{1, 2, 3\}$), explicitly ignoring the massive class imbalance caused by the background ($c = 0$):

$$\text{Foreground mIoU} = \frac{1}{3} \sum_{c=1}^3 \frac{|P_c \cap T_c|}{|P_c \cup T_c|}$$

where P_c and T_c are the sets of predicted and target pixels for class c , respectively.

While class-weighted accuracy and pixel-level F1 both handle class imbalance, neither captures exact geometric alignment as strictly as the Jaccard Index: accuracy underpenalizes structural hallucinations, and F1 is more forgiving of spatial misalignments. Since generative distillation demands precise reproduction of stroke boundaries and spatial coherence, Foreground mIoU is the most rigorous measure of structural fidelity.

4.2. Distillation Strategies

We test five distinct strategies for synthesizing the transfer batch to systematically isolate the cause of optimization failure.

1. Standard CAKE (Unbounded): Following the exact implementation of the original CAKE paper, we initialize the synthetic batch $X \sim \mathcal{N}(0, I)$. The pixel-wise target map Y is sampled from a uniform distribution over the four classes. The input X is then optimized via gradient descent to match Y .

2. Bounded CAKE: A variant of the standard approach where the input initialization is restricted to a bounded Gaussian-like prior in the range $[0, 1]$. This attempts to keep the initial noise closer to a standard image domain before optimization.

3. Shuffled Manifold Targets: To test whether the failure stems purely from unstructured targets, we initialize $X \sim \mathcal{N}(0, I)$ and perform a single forward pass through the teacher to generate a batch of valid, on-manifold output maps Y_{batch} . We then shuffle these valid maps across the batch to use as optimization targets for X .

4. Shifted Manifold Targets: Similar to the shuffled approach, we generate valid output maps from the teacher. Instead of shuffling across the batch, we apply a spatial shift of two pixels to the output map. This preserves local stroke coherence while forcing the optimization to attempt a slight geometric translation.

5. Noise Forward Pass ($N = 1$): Diverging entirely from boundary-seeking optimization, this strategy performs zero gradient updates on the input space. We sample $X \sim \mathcal{N}(0, I)$ and simply pass it through the teacher autoencoder. The resulting output is directly used as the target for the student. We found that a single iteration ($N = 1$) provides the most effective manifold projection.

4.3. Results and Analysis

Table 1 summarizes the performance of the student autoencoder across all five synthesis strategies.

Table 1: Student Autoencoder Fidelity by Synthesis Strategy. Foreground mIoU is the primary benchmark for spatial feature learning.

Synthesis Strategy	Overall Acc.	F-mIoU
Standard CAKE	0.8149	0.0345
Bounded CAKE	0.7159	0.0789
Shuffled Targets	0.8178	0.0038
Shifted Targets	0.8174	0.0070
Noise Forward Pass	0.8500	0.1891

Both CAKE variants fail to transfer geometric knowledge. Standard CAKE yields a near-zero Foreground mIoU of 0.0345, and bounding the initialization provides only marginal improvement (0.0789), leaving the fundamental gradient conflict unresolved.

Counterintuitively, providing valid manifold targets (Experiments 3 and 4) produces the worst results overall. Though shuffled and shifted targets are geometrically coherent, they are arbitrarily paired with unrelated input noise, forcing the student to learn a fractured, discontinuous mapping. The high directional consistency of the resulting gradient signal (0.68) actively drives the latent representation into a confident but geometrically mismatched state, causing catastrophic mode collapse. This shows that structured, mismatched targets are more destructive than pure noise — not less.

Direct noise propagation through the teacher decisively outperforms all other strategies with a Foreground mIoU of 0.1891. By passing input noise directly through the teacher without any gradient optimization, it sidesteps gradient conflicts entirely. Crucially, even off-distribution inputs produce outputs that preserve the teacher’s structural priors and learned Jacobian properties, giving the student smooth, functionally consistent training pairs rather than artificially decoupled ones.

5. CONCLUSION

In this work, we demonstrated that extending boundary-seeking distillation methods like CAKE to unsupervised generative models exposes a fundamental architectural incompatibility. Because an autoencoder’s dense output is strictly constrained by a low-dimensional latent bottleneck, independently sampled feature targets cause severe gradient conflicts across the learned manifold rather than discovering informative boundaries. To resolve this geometric impossibility, we find Noise Forward Pass, which abandons unstable gradient conflicts in favor of using the teacher model as a natural projector to generate safe, manifold-aware targets.

6. REFERENCES

- [1] Steven Braun, Martin Mundt, and Kristian Kersting, “Deep classifier mimicry without data access,” *Proceedings of Machine Learning Research*, vol. 238, pp. 4762–4770, 2024.
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